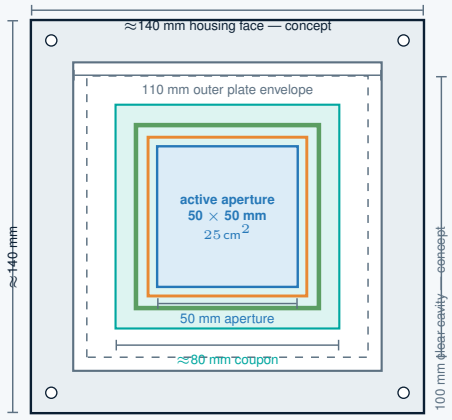
Integrated EO two-chamber cell — dimensioned concept views

Selected sealed two-chamber system with governing volume, membrane geometry and fabrication-release gates.

A Chamber face and central membrane cartridge

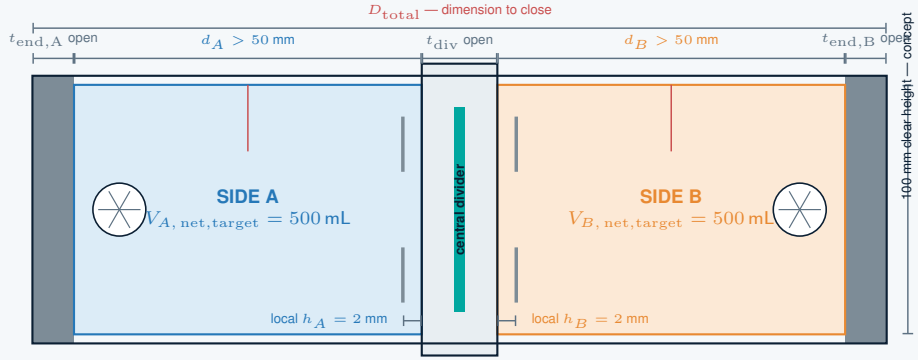


Nested face dimensions
Housing face: ≈140 × 140 mm [concept]
Clear chamber face: 100 × 100 mm [concept]
Outer plate envelope: 110 × 110 mm [starting]
Membrane coupon: ≈80 × 80 mm [starting]
Active aperture: 50 × 50 mm [fixed]

Radial lands on every edge
Housing edge → outer plate: 15 mm [concept]
Outer plate → coupon edge: 15 mm [starting]
Coupon edge → aperture: 15 mm [starting]

Within the 15 mm coupon land
3 mm inner seal + 2 mm vented guard groove +
3 mm outer seal + 7 mm spare land [starting]

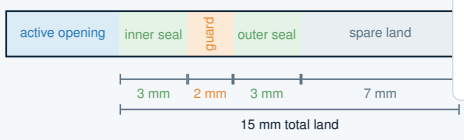
B Axial section and fabrication-release gates



Volume rule: A 100 × 100 × 50 mm empty cavity is 500 mL gross. Set d_A and d_B from measured fan, probe, baffle and port displacement, then calibrate each assembled chamber to 500 mL net free gas. Cavity depth remains release-gated.

C Seal-land chain and membrane-stack detail

Radial land chain: aperture → coupon



Sourced starting geometry
Maximum active aperture: 50 × 50 mm
Coupon: ≈80 × 80 mm
Outer plate envelope: 110 × 110 mm
Seal / guard / seal: 3 / 2 / 3 mm
Mask inserts: 40×40, 30×30, 20×20 and 8×50 mm

Stack through the active aperture — thickness exaggerated



Axial stack and local-gap status
Membrane thickness: 36.9 µm [model anchor]
Lot membrane thickness: open / measure
Porous-electrode thickness t_e: open / measure
Divider structure t_{div}: open / verify seal compression
Local membrane-face gap: 2 mm nominal [starting]
Interchangeable gap inserts: 1 / 2 / 4 mm
Active gap volume at 2 mm: 5.00 mL per side.

The local 2 mm membrane region contributes 5.00 mL to each 500 mL net free-gas target.

Release gates cover wall/divider sizing, port interfaces, fan, baffles, mounting, datums and tolerances.

Operating design state: 32 °C, initial $a_w = 0.50$, and membrane differential pressure target $|\Delta p| \leq 5$ Pa.

D Dimension and status register

Quantity	Value shown	Status
Net free gas target	500 ± 5 mL per side	Governing; calibration open
Chamber clear face	100 × 100 mm	Concept
Empty equivalent depth	50 mm per side	Gross-volume reference
Actual cavity depths	d_A, $d_B > 50$ mm	Dimension to close from displacement
Housing face	≈140 × 140 mm	Concept
Total axial depth	D_{total}	Dimension to close
Active aperture	50 × 50 mm = 25 cm²	Fixed design basis
Local membrane-face gap	2 mm nominal; 1 / 2 / 4 mm inserts	Sourced starting value
Active gap gas volume	5.00 mL per side at 2 mm	Subset of 500 mL net volume
Membrane coupon	≈80 × 80 mm	Sourced starting value
Outer plate envelope	110 × 110 mm	Sourced starting value
Inner seal / guard / outer seal	3 / 2 / 3 mm	Sourced starting value
Membrane thickness	36.9 µm	Model anchor; measure actual
Insulating frame thickness	8–10 mm	Starting value
Load-plate thickness	12 mm metal / ≥16 mm PEEK	Starting value; structural gate
End walls and divider	$t_{\text{end,A/B}}$, t_{div}	Open structural gate
Fluidic/electrical ports	Starting bores only	Open interface + leak gate
Internal fan	part, envelope, mount, heat	Open selection + test gate
Baffles	geometry and mounting	Open CAD + mixing gate
Fixture mounting/fasteners	interface, pattern, preload	Open structural gate
Datums and tolerances	drawing and inspection	Open release gate

Fabrication-release basis: Fixed and starting values define the engineering concept. The 500 mL value is a net free-gas target per side. Every required gate must close with cited evidence.

Engineering concept; the gate register defines the evidence required for fabrication release.